

Practical 12 :

// Import Spark libraries

import org.apache.spark.sql.SparkSession

object SimpleSparkApp {

def main(args: Array[String]): Unit = {

// Create Spark session

val spark = SparkSession.builder()

.appName("Simple Spark Scala Program")

.master("local[*]") // Run locally

.getOrCreate()

// Create sample data

val data = Seq(

(1, "Amit", 85),

(2, "Neha", 90),

(3, "Rahul", 78),

(4, "Sneha", 88)

)

// Convert to DataFrame

val df = spark.createDataFrame(data)

.toDF("id", "name", "marks")

// Show data

df.show()

// Filter students with marks > 80

val filteredDF = df.filter("marks > 80")

filteredDF.show()

// Calculate average marks

df.createOrReplaceTempView("students")

val avgMarks = spark.sql("SELECT AVG(marks) AS average_marks FROM students")

```
avgMarks.show()
```

```
// Stop Spark session
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```
spark.stop()
```

```
}
```

```
}
```